

Mohammad Abbasi

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Experience

Faragostar | AI Engineer

Apr 2025 – Present (Hybrid) | Tehran, Iran

- Built a letter-authoring chatbot for Faragostar's correspondence-automation system using n8n.
- Developed an n8n-based RAG chatbot for intelligent document search.
- Built a query-optimization system to modernize the legacy search workflow using n8n.
- Developed an n8n-based automated customer-response system.
- Developed and optimized Persian RAG systems for document retrieval using LangChain and LangGraph.
- Implemented, improved, and deployed enterprise-scale OCR, Summarizer, Feature Extraction, and Letter Writer services.
- Designed and improved Text-to-Speech and Speech-to-Text services through model fine-tuning and prompt optimization.
- Integrated and deployed language models (Qwen, Aya, Gemma, LLaMA) on Docker, Ollama, and VLLM infrastructure.
- Designed and implemented AgentCore and Agentic Systems modules with Long-Term Memory and interactive graphs.
- Managed and optimized infrastructure, including Docker networking, Kong API Gateway, PostgreSQL, and SQL Server.
- Fine-tuned language and speech models (Meta TTS and ASR) and designed training and evaluation pipelines.
- Fully documented services, managed Git repositories, and cleaned up codebases and documentation.

Malek Ashtar University | Researcher

Feb 2025 – Present (Part-time) | Tehran, Iran

- Satellite image processing (SAR/ISAR).

Demis | AI Specialist

Feb 2025 – Apr 2025 | Isfahan, Iran

- Speech-to-text project.
- AI-powered medical assistant project.

Entekhab Industrial Group | AI Specialist

Nov 2023 – Apr 2024 | Isfahan, Iran, SnowaTec

- RAG for instruction manuals.
- Food recommender system using a Telegram bot (available on GitHub).
- Feature extraction and named-entity recognition through language-model fine-tuning.
- Text generation and feature extraction using large language models.
- Built translation models using Transformers (available on GitHub).
- Developed a food recommender system based on food ingredients.
- Used ASR, TTS, large language models, and generative AI with NVIDIA™NEMO.
- Gathered and analyzed user comments about Digikala and Amazon products (available on GitHub).
- ...

Entekhab Industrial Group | Technology Monitoring Specialist

Sep 2022 – Nov 2023 | Isfahan, Iran, SnowaTec

- Monitored the artificial intelligence centers of household-appliance companies (Bosch, Samsung, LG, etc.).
- Examined the structure of AI platforms and online services of leading home-appliance companies.
- Conducted patent searches.
- Benchmarked smart home appliances and future trends.
- Benchmarked products in the home-appliance sector.
- Monitored enterprise AI tools.
- ...

Skills

PROGRAMMING

Python and IPython Notebook (machine learning and deep learning)
 R (beginner)
 JavaScript, HTML, and CSS (previously used)
 MySQL (previously used)

MACHINE LEARNING

Regression
 Cluster analysis
 Classification

DEEP LEARNING

Neural networks
 CNN
 GNN
 RNN (beginner)
 GANs (beginner)
 Encoder and decoder
 Transformers (beginner)
 ...

SOFTWARE

RapidMiner
 CVAT
 Label Studio
 LM Studio
 n8n
 L^AT_EX
 ...

OPERATING SYSTEMS

Linux (expert)
 Windows

PLATFORMS

Anaconda
 NVIDIA™NEMO
 Git
 GitHub
 Docker
 LibRecommender (recommender systems)
 Cornac (recommender systems)

PRODUCTIVITY SOFTWARE

Microsoft Office (Excel, Word, PowerPoint)
 Canva
 Adobe Photoshop
 Adobe Audition
 Adobe Premiere

Iran Smartech | Content Creator and Website Designer
2021 – 2022 | Isfahan, Iran

- Designed elements and posters with Photoshop.
- Developed websites using WordPress.

Farda Sanjesh Sarmayeh (FSSco.) | Website Designer
2015 – 2016 | Isfahan, Iran

- Developed websites using WordPress.

Education

Tarbiat Modares University | PhD in Artificial Intelligence
2025 – Present | Tehran, Iran

- Completed courses in Generative Artificial Intelligence, Complex Networks, Big Data Analytics, Artificial Intelligence Seminar, and Advanced Algorithms.

University of Isfahan | MSc in Artificial Intelligence
2022 – 2024 | Isfahan, Iran

- Advanced algorithms, advanced theory of computation, advanced artificial intelligence, machine learning, theoretical foundations of cryptography, computational data mining, and deep learning.
- Healthy food recommender system using users' dietary-habit data mining (thesis).
- Persian speech-to-speech chatbot (available on GitHub).
- Big-data recommender system using Two-Tower, DeepFM, and related models.
- Healthy-food recommender system based on nutritional knowledge.
- Review of food recommender systems.
- Big-data preprocessing for deep-learning and machine-learning applications.
- ...

University of Isfahan | BSc in Computer Science
2017 – 2022 | Isfahan, Iran

- Computer basics and programming, advanced programming, mathematical foundations, general mathematics 1 and 2, principles of computer systems, combinatorics, matrices and linear algebra, logic and set theory, numerical analysis, operating systems, design and analysis of algorithms, databases, probability, differential equations, linear optimization, theory of computation, algebra, cryptography, coding theory, LaTeX, MATLAB, special topics in computer science (artificial intelligence), and logic circuits.

Honors and Publications

- Graduated in the top 15% of the BSc cohort – University of Isfahan.
- Healthy food recommender system via users' dietary-habit data mining – 15th International Conference on Information and Knowledge Technology (IKT).

Courses Completed

- Git – Great Learning (certified).
- LLM Hackathon – Quera (certified).
- Data Science Foundations: Data Mining – Lynda.
- R Programming A-Z: R for Data Science with Real Exercises – Udemy.
- Learn the 2020 Advanced Python Programming – Udemy.
- CS50AI 2023 – Harvard University.

Books Read

- Data Mining: Concepts and Techniques (Jiawei Han).
- Introduction to Algorithms (CLRS).
- Artificial Intelligence: A Modern Approach (2010, Prentice Hall).
- Decision Support Systems for Business Intelligence, 2nd edition (2010, Wiley).
- Matrix Methods in Data Mining and Pattern Recognition (Lars Eldén).
- Deep Learning with Python (François Chollet).

Interests

RESEARCH INTERESTS

Large language models
Recommender systems
SAR and ISAR image processing
AI futurology
AI policy-making
Industry 4.0

RECENT TECHNOLOGY STACK

Languages and Frameworks: Python (FastAPI, Flask, Streamlit), JavaScript (Node.js), HTML/CSS, Bash Script

Infrastructure and Deployment: Docker, Docker Compose, Docker Swarm, Ollama, VLLM, Llama Factory, Kong API Gateway, n8n, DEC, Nginx, Apache Server, Cloudflare, SQL Server, PostgreSQL, pgAdmin

Machine Learning and LLMs: LangChain, LangGraph, Hugging Face Transformers, PyTorch, TensorFlow, NVIDIA NEMO, Sentence Transformers, Whisper, Qwen, Aya, LLaMA, Gemma, Mistral, GPT API, Cohere

Image and Text Processing: OpenCV, Tesseract OCR, OCR Pipelines, Feature Extraction, Summarizer, Validation Services, Synonym Generator

Data Management and Model Training: Fine-tuning Pipelines, PEFT, Flash Attention, Data Preprocessing (ASR, OCR, TTS), Anaconda, Pandas, NumPy

Version Control and Documentation: Git, GitHub, Markdown, LaTeX, structure.md, README Optimization

Development and Testing Tools: VS Code, Postman, API Explorer, Gradio, Flake8, LM Studio

TEXT MODELS USED

Qwen 2.5
Qwen 3
Cohere Aya
Llama 3
MT5
Sentence-transformer base models

TEXT + IMAGE MODELS USED

Qwen 2.5 VL
Mistral

HOBBIES

Photography
Cycling
Running
Theater
Cinema

LANGUAGES

Persian (native)
English (fluent)